The type of quasi-inverses of a function with an inhabited quasi-inverse is the type of self-homotopies of the identity on A

The Univalent Foundations Program

hott-book:equivalences.tex:local-snapshot-20261011; CC BY-SA 3.0 Unported

Editorial note. The counted statement is the author's 'lem:qinv-autohtpy' (source lines 42-45); the proof environment starts on the next line, so the pairing is by adjacency.

Editorial note. No context span is retained: the statement and proof are self-contained apart from three labelled invocations, which are declared as external navigation pointers.

Editorial note. The retained counter values make the counted result print with its original book number, and the chapter and section counters are set so that the retained context (where present) prints with its own original number as well.

Editorial note. The original number of the counted result was inferred by counting the theorem-class environments of its section of the pinned source, with the two labelled neighbours of that section matching the same count; it is the corresponding entry of the same count, not a bibliography entry.

Editorial note. This source directory intentionally contains only the author file, so the retained book macro library is bundled as a per-task copy inside this task directory and referenced from preamble_spans; the copy is verbatim from the same pinned book snapshot and no macro is added, changed or invented.

Editorial scope. One original result of the HoTT Book, counted once: the quasi-inverses of a function with inhabited quasi-inverse are the self-homotopies of the identity (author label lem:qinv-autohtpy).

Allowed standard prerequisites (editorial).

- Quasi-inverse. qinv(f) is the type of pairs (g, g o f ~ id) x (f o g ~ id).. It is the type whose equivalence with the self-homotopies of the identity is asserted.
- Univalence and idtoeqv. idtoeqv : (A = B) -> (A ~ B) is an equivalence, so by univalence any equivalence may be assumed to be of the form idtoeqv(p).. It is the reduction performed at source line 48.
- Function extensionality. Homotopies may be replaced by equalities of functions.. It is used twice, at source lines 55-56 and at line 62.

Source reference. Original exercise 2.10: sources/hott-hits/book-labelnumbers-first-edition.tex, source lines 1402–1402. Original result cited by the retained text but located outside the retained span. It is a navigation pointer only: the counted statement's own author proof is retained line for line and no part of it is replaced by this reference. The reference number is taken from the label-number dump of the pinned book snapshot.

Source reference. Original theorem 3.11.8: sources/hott-hits/book-labelnumbers-first-edition.tex, source lines 1760–1760. Original result cited by the retained text but located outside the retained span. It is a navigation pointer only: the counted statement's own author proof is retained line for line and no part of it is replaced by this reference. The reference number is taken from the label-number dump of the pinned book snapshot.

Source reference. Original theorem 3.11.9: sources/hott-hits/book-labelnumbers-first-edition.tex, source lines 1844–1844. Original result cited by the retained text but located outside the retained span. It is a navigation pointer only: the counted statement's own author proof is retained line for line and no part of it is replaced by this reference. The reference number is taken from the label-number dump of the pinned book snapshot.

Lemma 4.1.1. *If $f : A \to B$ is such that $\mathsf{qinv}(f)$ is inhabited, then*

$$\mathsf{qinv}(f) \simeq \Big(\prod_{x:A} (x = x)\Big).$$

Proof. By assumption, f is an equivalence; that is, we have $e : \mathsf{isequiv}(f)$ and so $(f, e) : A \simeq B$. By univalence, $\mathsf{idtoeqv} : (A = B) \to (A \simeq B)$ is an equivalence, so we may assume that (f, e) is of the form $\mathsf{idtoeqv}(p)$ for some $p : A = B$. Then by path induction, we may assume p is refl_A, in which case f is id_A. Thus we are reduced to proving $\mathsf{qinv}(\mathsf{id}_A) \simeq (\prod_{(x:A)}(x = x))$. Now by definition we have

$$\mathsf{qinv}(\mathsf{id}_A) \equiv \sum_{g:A\to A} \big((g \sim \mathsf{id}_A) \times (g \sim \mathsf{id}_A)\big).$$

By function extensionality, this is equivalent to

$$\sum_{g:A\to A} \big((g = \mathsf{id}_A) \times (g = \mathsf{id}_A)\big).$$

And by original exercise 2.10, this is equivalent to

$$\sum_{h:\sum_{(g:A\to A)}(g=\mathsf{id}_A)} (\mathsf{pr}_1(h) = \mathsf{id}_A)$$

However, by original theorem 3.11.8, $\sum_{(g:A\to A)}(g = \mathsf{id}_A)$ is contractible with center $(\mathsf{id}_A, \mathsf{refl}_{\mathsf{id}_A})$; therefore by original theorem 3.11.9 this type is equivalent to $\mathsf{id}_A = \mathsf{id}_A$. And by function extensionality, $\mathsf{id}_A = \mathsf{id}_A$ is equivalent to $\prod_{(x:A)} x = x$. □